UNIT 2 – MODULES AND PACKAGES

Advanced Python Programming (DI02016011)
Diploma in Engineering – Information Technology

Topic 2.1: Introduction to Module

Definition:
A **module** in Python is a file containing Python code such as functions, variables, and classes that can be reused in another Python program.

Explanation:
A module helps us divide a large program into smaller and manageable files. Instead of writing the same code again and again, we can create it once in a module and use it whenever required.

For example, a file named `calculator.py` can contain mathematical functions.

Example:

```
def add(a, b):
    return a + b
```

This function can later be used in another Python program by importing the module.

3 Uses: To reuse Python code.To divide a large program into smaller parts.To make programs easier to manage. **3 Features:**Modules can contain functions and variables.Modules can be imported into other programs.They support code reusability. **Applications:**Large software projectsCalculator programsData processing programs **Real-Life Example:**
In a banking application, separate modules can be created for **account management, payments, and customer information**.

Topic 2.2: Creating User-Defined Module

Definition:
A **user-defined module** is a Python file created by the programmer to store reusable Python code.

Explanation:
Python allows programmers to create their own modules according to their requirements. A module is simply saved with the `.py` extension.

For example, we can create a module named `student.py`.

Example:

```
name = "Rahul"

def display():
    print("Student Name:", name)
```

The `student.py` file can then be used in another Python program.

3 Uses: To store frequently used functions.To organize program code.To reuse code in different

programs. **3 Features:**Created by the programmer.Saved with `.py` extension.Can contain variables, functions and classes. **Applications:**Student management systemsBilling systemsBusiness applications **Real-Life Example:**
A college project can have a `student.py` module containing functions for **adding, displaying and searching student records**.

Topic 2.3: Importing a Module in Python

Definition:
Importing a module means making the code of a module available for use in another Python program.

Explanation:
Python provides different ways to import modules. The GTU syllabus includes **normal import, from import, and from import with *.**

Example – Normal Import:

```
import math

print(math.sqrt(25))
```

Example – From Import:

```
from math import sqrt

print(sqrt(25))
```

Example – From Import with *:

```
from math import *

print(sqrt(25))
```

3 Uses: To use functions from another module.To reuse existing Python code.To reduce repeated programming. **3 Features:**Supports different import methods.Makes module functions available.Helps in code reuse. **Applications:**Mathematical calculationsFile processingData analysis **Real-Life Example:**
A student result program can import a calculation module to calculate **total marks and percentage**.

Topic 2.4: Module Search Path

Definition:
The **module search path** is the list of locations where Python looks for a module when we use the `import` statement.

Explanation:
When a module is imported, Python searches different locations to find that module. If Python cannot find the required module in the available search locations, it generates an import error.

The module search path helps Python identify where modules are stored.

3 Uses: Helps Python find modules.Allows modules to be stored in different locations.Supports importing user-defined modules. **3 Features:**Used during module importing.Contains locations searched by Python.Helps locate required Python files. **Applications:**Large Python

projectsUser-defined modulesPackage management **Real-Life Example:**
Suppose a project has a separate folder containing calculation modules. The module search path helps Python locate those modules when they are imported.

Topic 2.5: OS Module

Definition:
The **OS module** in Python provides functions for interacting with the operating system, such as working with directories and files.

Explanation:
The GTU syllabus includes working with the **current working directory, creating a directory, and listing files and directories** using the OS module.

Example:

```
import os

print(os.getcwd())
```

`os.getcwd()` returns the current working directory.

3 Uses: To work with directories.To find the current working directory.To list files and folders. **3 Features:**Provides operating-system related functions.Can create and manage directories.Can access file and directory information. **Applications:**File managementDirectory managementAutomation programs **Real-Life Example:**
A backup program can use the OS module to find a folder and manage the files stored inside it.

Topic 2.6: Introduction to Packages

Definition:
A **package** in Python is a collection of related modules organized together in a directory.

Explanation:
Packages help organize large Python programs. Instead of keeping many modules in one place, related modules can be grouped inside a package. This makes the program easier to understand and maintain.

3 Uses: To organize related modules.To manage large Python projects.To improve code organization. **3 Features:**Contains related modules.Provides better program structure.Supports reusable code. **Applications:**Large software projectsWeb applicationsBusiness applications **Real-Life Example:**
A college management system can have a package containing separate modules for students, teachers, fees and attendance.

Topic 2.7: Creating User-Defined Package

Definition:
A **user-defined package** is a package created by a programmer to organize their own related Python modules.

Explanation:
A programmer can create a directory and place related Python modules inside it. This provides a clear

structure for a large project.

3 Uses: To group related modules.To organize project files.To reuse modules in different programs. **3 Features:**Created according to project requirements.Can contain multiple modules.Makes code easier to manage. **Applications:**College management systemsBanking softwareLarge Python projects
Real-Life Example:
A school software project can create a package containing modules for students, teachers and examination records.

Topic 2.8: Importing a Package in Python

Definition:
Importing a package means using the modules or functions stored inside a Python package in another program.

Explanation:
The GTU syllabus includes **normal import, from import, and from import with *** for importing packages.

3 Uses: To use modules stored in a package.To reuse organized Python code.To access required functions. **3 Features:**Supports different import methods.Allows access to package modules.Improves code reuse. **Applications:**Large applicationsData processingWeb development **Real-Life Example:**
A billing application can import a package containing modules for customer details, products and invoice calculations.

Topic 2.9: Intra-Package References

Definition:
An **intra-package reference** is a reference used by one module inside a package to access another module within the same package.

Explanation:
In a large package, modules may need to use functions or data from other modules in that same package. Intra-package references help these modules work together.

3 Uses: To connect modules inside a package.To reuse functions between related modules.To organize large applications. **3 Features:**Works between modules of a package.Supports modular programming.Reduces repeated code. **Applications:**Large Python applicationsBusiness softwareManagement systems **Real-Life Example:**
Inside a college package, an attendance module can use functions from a student-information module.

Topic 2.10: Installing PIP

Definition:
PIP is a package management tool used in Python to install and manage Python packages.

Explanation:
PIP makes it easier to obtain Python packages required for a project. It is commonly used from the command line.

Example:

```
pip install package_name
```

3 Uses: To install Python packages.To manage project dependencies.To obtain additional Python libraries. **3 Features:**Works from the command line.Can install external packages.Helps manage Python dependencies. **Applications:**Data science projectsWeb developmentMachine learning projects
Real-Life Example:
A programmer developing a data analysis application can use PIP to install a required data-processing package.

Topic 2.11: Installing and Uninstalling Python Packages

Definition:
Installing a Python package means adding a package to the Python environment, while uninstalling means removing it when it is no longer required.

Example – Install:

```
pip install package_name
```

Example – Uninstall:

```
pip uninstall package_name
```

3 Uses: Install required libraries.Remove unnecessary packages.Manage project requirements. **3 Features:**Packages can be installed using PIP.Packages can be removed using PIP.Helps maintain the Python environment. **Applications:**Software developmentWeb applicationsData analysis projects
Real-Life Example:
If a developer needs a library for a project, it can be installed with PIP. After the project is completed, an unnecessary package can be uninstalled.